

HYDROGENATED NBR: CATALYTIC HYDROGENATION, SIMULTANEOUS METATHESIS-HYDROGENATION, AND CATALYST-RESIDUE CONTROL PATENTS

1. ABSTRACT

This report synthesizes five primary patents relevant to hydrogenated nitrile rubber (HNBR) preparation, with emphasis on routes that reduce residual unsaturation, control molecular weight, and improve processability or product stability. The target material is HNBR derived from nitrile rubber (NBR), typically a copolymer of a conjugated diene and an α,β -unsaturated nitrile, with hydrogenation used to improve heat resistance, oil resistance, and oxidative stability while preserving the nitrile rubber microstructure. The selected patents span two major process families: (1) simultaneous hydrogenation and metathesis of NBR using ruthenium/osmium-based catalysts to lower molecular weight and narrow molecular weight distribution in a single step, and (2) conventional hydrogenation of NBR using noble-metal catalysts followed by catalyst-removal or catalyst-management strategies to improve aging performance and product quality.

Across the landscape, the metathesis-hydrogenation patents (US7919563B2, US9150669B2, EP2473280B1) focus on producing low-molecular-weight HNBR with reduced Mooney viscosity and narrower polydispersity. They disclose catalyst families based on Ru/Os complexes, optional co-olefins, hydrogen atmosphere, and reaction temperatures around 80 degrees C in the examples, with catalyst loadings reported in phr and metal contents in ppm. The examples show that catalyst identity strongly affects hydrogenation rate, residual double bond (RDB) decay, and final Mooney viscosity. In contrast, US7846995B2 and EP3827029A1 address downstream catalyst residue control and direct hydrogenation quality, respectively. US7846995B2 uses functionalized ion-exchange resin treatment to remove ruthenium-containing residues from optionally hydrogenated nitrile rubber solutions, while EP3827029A1 defines hydrogenation windows for Ru, Pd, or Rh catalysis and links low catalyst residue to improved aging, reduced discoloration, and lower gel growth.

Overall, the evidence supports HNBR synthesis by selective hydrogenation of NBR in solution, with optional simultaneous chain scission via metathesis for viscosity reduction, followed by rigorous catalyst-removal or residue-minimization steps to preserve aging stability and appearance.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: US7919563B2
- Patent Title: Low molecular weight hydrogenated nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2011
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses a two-step concept in which nitrile rubber is subjected to metathesis in the presence of Ru or Os catalysts, optionally with a co-olefin and solvent, to generate hydrogenated nitrile rubber with lower molecular weight and narrower molecular weight distribution. The process is positioned as a route to low-Mooney HNBR with improved processability while retaining the HNBR microstructure. - Molecular weight (MW): 30,000 to 250,000 — Claim 1 - Mooney viscosity (ML 1+4 @ 100 deg. C.): 3 to 50 — Claim 1 - MWD / polydispersity index: less than 2.5 — Claim 1 - Co-olefin pressure: 500 psi — Table 1 - Reactor temperature: 80 degrees C. — Table 1 - Catalyst load: 0.05 phr — Table 1 - Catalyst load: 0.10 phr — Table 1 - Catalyst load: 0.25 phr — Table 1 - Metal content: 65.6 ppm — Table 1 - Solid concentration: 15 wt.-% — Example 1 - Solid concentration: 6 wt.-% — Example 3 - Ethylene pressure: 6.9×10^5 Pa — Example 1 - Ethylene pressure: 6.9×10^5 Pa — Example 2 - Ethylene pressure: 6.9×10^5 Pa — Example 3 - 1-hexene amount: 150 g — Example 4 - Nitrogen degassing cycles: 3 — Example 4 - Mn: 27 kg/mol — Table 2 - Mw: 54 kg/mol — Table 2 - PDI: 2.0 — Table 2 - Mooney viscosity: 3 — Table 2 - Mn: 85 kg/mol — Substrate - Mw: 296 kg/mol — Substrate - PDI: 3.4 — Substrate

Relevant Experimental Evidence - Example 1: Ethylene-assisted metathesis/hydrogenation at 80 degrees C with 0.05 phr catalyst gave reduced molecular weight and lower Mooney viscosity. - Example 2: Increasing catalyst loading to 0.10 phr maintained the same process framework and produced a lower-Mooney product than the starting polymer. - Example 3: Lower solids concentration with ethylene still produced substantial molecular-weight reduction and narrower PDI. - Example 4: Using 1-hexene as co-olefin at 80 degrees C demonstrated that co-olefin choice affects final Mn, Mw, PDI, and Mooney viscosity. - Example 4: The product showed Mn 27 kg/mol, Mw 54 kg/mol, and PDI 2.0, illustrating strong depolymerization control.

Technical Relevance This patent is highly relevant because it directly targets low-molecular-weight HNBR, a key need for improved processability. It shows that simultaneous

metathesis/hydrogenation can tune residual unsaturation and viscosity in one operation.

Patent 2

- Patent Number: US9150669B2
- Patent Title: Process for the preparation of low molecular weight hydrogenated nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2015
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses simultaneous hydrogenation and metathesis of nitrile rubber in the presence of hydrogen and a Ru/Os catalyst system, optionally with a co-olefin, to produce low-molecular-weight HNBR in a single step. The emphasis is on catalyst selection that improves hydrogenation efficiency while also reducing molecular weight and narrowing distribution. - Catalyst amount: 0.041 phr — Table 1 - Catalyst amount: 0.045 phr — Table 1 - Catalyst amount: 0.060 phr — Table 1 - Metal content: 65.6 ppm — Table 1 - Co-catalyst amount: 1.0 phr — Table 1 - Reaction time: 0 min — Table 2 - Reaction time: 60 min — Table 2 - Reaction time: 120 min — Table 2 - Reaction time: 180 min — Table 2 - Reaction time: 240 min — Table 2 - RDB: 100% at 0 min — Table 2 - RDB: 2.5% at 60 min — Table 2 - RDB: 0.6% at 120 min — Table 2 - RDB: 0.5% at 180 min — Table 2 - RDB: 0.4% at 240 min — Table 2 - RDB: 7.0% at 60 min — Table 2 - RDB: 4.5% at 120 min — Table 2 - RDB: 3.9% at 180 min — Table 2 - RDB: 3.5% at 240 min — Table 2 - RDB: 15.4% at 60 min — Table 2 - RDB: 3.1% at 120 min — Table 2 - RDB: 1.3% at 180 min — Table 2 - RDB: 0.7% at 240 min — Table 2 - Mw: 113371 g/mol — Table 3 - Mn: 48352 g/mol — Table 3 - PDI: 2.34 — Table 3 - Mooney viscosity: 33.3 — Table 3 - Mw: 133758 g/mol — Table 3 - Mn: 54595 g/mol — Table 3 - PDI: 2.45 — Table 3 - Mooney viscosity: 41.3 — Table 3 - Mw: 195304 g/mol — Table 3 - Mn: 64130 g/mol — Table 3 - PDI: 3.05 — Table 3 - Mooney viscosity: 69.3 — Table 3

Relevant Experimental Evidence - Example 1: Three catalyst systems were compared for hydrogenation rate and final polymer properties. - Example 1: The Hoveyda catalyst system achieved the fastest RDB reduction, reaching 0.4% at 240 min. - Example 1: The inventive catalyst gave lower Mw and Mooney viscosity than the industrial benchmark. - Example 1: The catalyst with triphenylphosphine co-catalyst showed slower hydrogenation and higher final Mooney viscosity. - Example 1: The data demonstrate that catalyst choice controls both hydrogenation kinetics and molecular-weight reduction.

Technical Relevance This patent is directly relevant because it couples hydrogenation with metathesis to make low-MW HNBR efficiently. It also provides comparative evidence that catalyst system selection strongly affects residual unsaturation and rheology.

Patent 3

- Patent Number: EP2473280B1
- Patent Title: Process for the preparation of hydrogenated nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2014
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses simultaneous hydrogenation and metathesis of nitrile rubber using Ru/Os catalysts of defined general formulae, optionally with a co-olefin, to obtain HNBR with lower molecular weight and narrower molecular weight distribution. The examples compare Grubbs III, Grubbs II, and Wilkinson's catalyst, showing that catalyst identity influences hydrogenation rate and final viscosity. - Catalyst amount: 0,057 phr — Example 1 - Catalyst amount: 0,080 phr — Example 1 - Catalyst amount: 0,06 phr — Example 1 - Catalyst amount: 0,055 phr — Example 1 - Metal content: 65,6 ppm — Example 1 - Metal content: 91,3 ppm — Example 1 - Reaction time: 0 min — Example 1 - Reaction time: 60 min — Example 1 - Reaction time: 120 min — Example 1 - Reaction time: 180 min — Example 1 - Reaction time: 240 min — Example 1 - RDB: 100% at 0 min — Example 1 - RDB: 2,0% at 60 min — Example 1 - RDB: 0,7% at 60 min — Example 1 - RDB: 15,4% at 60 min — Example 1 - RDB: 2,9% at 60 min — Example 1 - RDB: 0,4% at 120 min — Example 1 - RDB: 0,5% at 120 min — Example 1 - RDB: 3,1% at 120 min — Example 1 - RDB: 0,6% at 120 min — Example 1 - RDB: 0,3% at 180 min — Example 1 - RDB: 0,3% at 180 min — Example 1 - RDB: 1,3% at 180 min — Example 1 - RDB: 0,3% at 180 min — Example 1 - RDB: 0,6% at 240 min — Example 1 - Mw: 191541 g/mol — Example 1 - Mn: 68643 g/mol — Example 1 - PDI: 2,79 — Example 1 - Mooney viscosity: 38,2 — Example 1 - Mw: 135772 g/mol — Example 1 - Mn: 58259 g/mol — Example 1 - PDI: 2,33 — Example 1 - Mooney viscosity: 20,2 — Example 1 - Mw: 195304 g/mol — Example 1 - Mn: 64130 g/mol — Example 1 - PDI: 3,05 — Example 1 - Mooney viscosity: 69,3 — Example 1 - Mw: 144212 g/mol — Example 1 - Mn: 59851 g/mol — Example 1 - PDI: 2,45 — Example 1 - Mooney viscosity: 32,6 — Example 1

Relevant Experimental Evidence - Example 1: Grubbs III, Grubbs II, and Wilkinson's catalyst were compared at similar metal contents. - Example 1: Grubbs III hydrogenated faster in the first 120 min than Grubbs II or Wilkinson's catalyst. - Example 1: The catalyst systems produced HNBR with different Mw, PDI, and Mooney viscosity values. - Example 1: The lowest Mooney viscosity in the table was 20.2, showing strong processability improvement. - Example 1: The example demonstrates that catalyst choice can tune both hydrogenation kinetics and final rheology.

Technical Relevance This patent is relevant because it provides a practical catalyst-comparison framework for simultaneous hydrogenation/metathesis of NBR. It supports the idea that Ru-based catalyst design is central to controlling residual unsaturation and molecular weight in HNBR.

Patent 4

- Patent Number: US7846995B2
- Patent Title: ... -containing catalyst residues from optionally hydrogenated nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2010
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent focuses on post-hydrogenation purification: removing ruthenium-containing catalyst residues from optionally hydrogenated nitrile rubber solutions using functionalized ion-exchange resins. The process is intended to improve product appearance and aging behavior by lowering residual noble-metal content after metathesis or hydrogenation processing. - Resin mass: 0.5 g — Example 2A - Rubber solution mass: 180 g — Example 2A - Flask volume: 500 ml — Example 2A - Reaction temperature: ca. 100 degrees C — Example 2A - Reaction time: 66 hours — Example 2A - Drying temperature: 60 degrees C — Example 2A - Initial Ru content: 15 ppm — Table 2 - Inventive process Ru content: 10 ppm — Table 2 - Ru removal: 33% — Table 2

Relevant Experimental Evidence - Example 2A: Treatment of nitrile rubber solution with Lewatit MonoPlus MP 500 reduced Ru content to 10 ppm. - Example 2A: The resin treatment was conducted at about 100 degrees C for 66 hours. - Example 2B: The untreated control established the baseline Ru content for comparison. - Example 2A: ICP analysis showed 33% Ru removal relative to the control. - Example 1: The example block indicates the process is aimed at catalyst-residue cleanup rather than polymer backbone modification.

Technical Relevance This patent is relevant because catalyst residue removal is critical for HNBR aging stability, color, and downstream performance. It complements hydrogenation patents by addressing the purification step needed after Ru-based synthesis.

Patent 5

- Patent Number: EP3827029A1
- Patent Title: Method for producing hydrogenated nitrile rubber and hnbr compositions thereof
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2021
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses a hydrogenation method for NBR in solution using ruthenium, palladium, or rhodium compounds, with the goal of

producing HNBR having good aging properties and low catalyst residue. The process emphasizes controlling noble-metal loading, hydrogenation temperature, pressure, and time to balance conversion, catalyst cost, and product quality. - Conjugated diene content: 40% to 90% by weight — Claim 1 - Alpha,beta-unsaturated nitrile content: 10% to 60% by weight — Claim 1 - Ruthenium amount: 10 ppm to 200 ppm — Claim 1 - Ruthenium amount: 10 ppm to 150 ppm — Claim 1 - Ruthenium amount: 10 ppm to 120 ppm — Claim 1 - Ruthenium amount: 10 ppm to 79 ppm — Claim 1 - Ruthenium amount: 42 ppm to 79 ppm — Claim 1 - Palladium amount: 20 ppm to <67 ppm — Claim 1 - Palladium amount: 44 ppm to <67 ppm — Claim 1 - Rhodium amount: 50 ppm to <270 ppm — Claim 1 - Rhodium amount: 79 ppm to <270 ppm — Claim 1 - Rhodium amount: 50 ppm to 250 ppm — Claim 1 - Rhodium amount: 50 ppm to 170 ppm — Claim 1 - Rhodium amount: 79 ppm to 170 ppm — Claim 1 - Hydrogenation temperature: 60 to 200 degrees C — Claim 1 - Hydrogenation pressure: 700000 pascals to 15000000 pascals — Claim 1 - Hydrogenation time: 1 to 24 hours — Claim 1

Relevant Experimental Evidence - No numbered examples were detected by the parser. - The claims define catalyst loading windows for Ru, Pd, and Rh systems. - The claims define hydrogenation temperature, pressure, and time windows. - The patent links low catalyst residue to improved aging, gel content, and appearance. - The disclosure is focused on direct HNBR production rather than post-treatment purification.

Technical Relevance This patent is relevant because it defines practical operating windows for hydrogenating NBR into HNBR while controlling noble-metal residue. It is especially useful for selecting catalyst loading and reaction conditions that preserve aging performance and product quality.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- US7919563B2, US9150669B2, and EP2473280B1 all target simultaneous hydrogenation/metathesis of NBR to reduce molecular weight and Mooney viscosity, but they differ in catalyst families and comparative performance data.
- US7919563B2 reports catalyst loadings of 0.05 to 0.25 phr and co-olefin use at 500 psi or 6.9×10^5 Pa, with 80 degrees C as the reactor temperature in the examples.
- US9150669B2 uses lower catalyst loadings of 0.041 to 0.060 phr and includes a 1.0 phr co-catalyst in one formulation; its tables show RDB dropping from 100% to as low as 0.4% by 240 min.
- EP2473280B1 reports catalyst loadings of 0.055 to 0.080 phr and metal contents of 65.6 to 91.3 ppm, with Grubbs III outperforming Grubbs II and Wilkinson's catalyst in early hydrogenation rate.
- Across the three metathesis-hydrogenation patents, final Mw values fall in the approximate 113,371 to 195,304 g/mol range in the examples, with Mooney viscosity spanning 3.0 to 69.3 depending on catalyst system and conditions.

- US7846995B2 is distinct because it does not modify the polymer backbone; instead it removes Ru residues from optionally hydrogenated nitrile rubber using ion-exchange resin at about 100 degrees C for 66 hours.
- EP3827029A1 shifts from catalyst choice to process windows, specifying 60 to 200 degrees C, 700000 to 15000000 pascals, and 1 to 24 hours for hydrogenation with Ru, Pd, or Rh catalysts.
- The combined evidence suggests that low residual unsaturation is best achieved by Ru-based hydrogenation/metathesis under moderate temperature and controlled catalyst loading, followed by residue-removal or residue-minimization steps to protect aging properties.

4. CONCLUSION

The patent set shows two complementary strategies for HNBR manufacture: one-step simultaneous hydrogenation/metathesis to lower molecular weight and viscosity, and post-reaction catalyst cleanup or catalyst-residue minimization to improve aging and appearance. The strongest synthesis evidence supports Ru-based catalyst systems, optional co-olefin use, and reaction temperatures near 80 degrees C for metathesis-assisted molecular-weight reduction, while EP3827029A1 defines broader hydrogenation windows of 60 to 200 degrees C, 700000 to 15000000 pascals, and 1 to 24 hours. For practical HNBR synthesis, the evidence favors catalyst loadings in the low phr range, careful control of metal ppm, and aggressive residue removal when noble-metal catalysts are used. The best-supported route is selective hydrogenation of NBR in solution, with simultaneous metathesis when low-MW HNBR is desired, followed by purification to reduce Ru/Rh/Pd residues and preserve aging stability.

5. REFERENCES

- US7919563B2 | Low molecular weight hydrogenated nitrile rubber | Not disclosed | US | 2011 | <https://patents.google.com/patent/US7919563B2/en>
- US9150669B2 | Process for the preparation of low molecular weight hydrogenated nitrile rubber | Not disclosed | US | 2015 | <https://patents.google.com/patent/US9150669B2/en>
- EP2473280B1 | Process for the preparation of hydrogenated nitrile rubber | Not disclosed | EP | 2014 | <https://patents.google.com/patent/EP2473280B1/en>
- US7846995B2 | ... -containing catalyst residues from optionally hydrogenated nitrile rubber | Not disclosed | US | 2010 | <https://patents.google.com/patent/US7846995B2/en>
- EP3827029A1 | Method for producing hydrogenated nitrile rubber and hnbr compositions thereof | Not disclosed | EP | 2021 | <https://patents.google.com/patent/EP3827029A1/en>